

Functions and Function Notation (Part III) Homework

Write the domain of each function below using interval notation. Then write the domain using set notation.

① $f(x) = \sqrt{3x}$

② $g(x) = \frac{2}{x+6}$

③ $h(x) = 7 - \log_2\left(\frac{x}{5} + 2\right)$

④ $f(x) = \begin{cases} 2x-1, & 3 < x < 5 \\ x+6, & 7 < x < 14 \end{cases}$

⑤ $g(x) = -\frac{2}{5x}$

⑥ $h(x) = \sqrt{8-x}$

⑦ $f(x) = \log_4\left(\frac{x}{2}\right)$

⑧ $g(x) = \begin{cases} 9^x, & -3 \leq x < -2 \\ x, & -2 < x < 10 \end{cases}$

⑨ $h(x) = \frac{7}{3-8x}$

⑩ $f(x) = \begin{cases} |x|, & -4 \leq x < -3 \\ -4, & -3 \leq x \leq -1 \end{cases}$

⑪ $g(x) = \begin{cases} x^2, & 3 \leq x < 8 \\ |2x+3|, & 9 < x < 10 \end{cases}$

⑫ $h(x) = \begin{cases} 5-7x, & -6 \leq x < 0 \\ x+6, & 4 \leq x \leq 7 \end{cases}$

⑬ $f(x) = 3x$

⑭ $g(x) = 5 - 2x^2$

⑮ $h(x) = 7^{x-6}$

Answers

- ① $[0, \infty)$, $\{x \mid x \geq 0\}$
- ② $(-\infty, -6) \cup (-6, \infty)$, $\{x \mid x \neq -6\}$
- ③ $(-10, \infty)$, $\{x \mid x > -10\}$
- ④ $(3, 5) \cup (7, 14)$, $\{x \mid 3 < x < 5 \text{ or } 7 < x < 14\}$
- ⑤ $(-\infty, 0) \cup (0, \infty)$, $\{x \mid x \neq 0\}$
- ⑥ $(-\infty, 8]$, $\{x \mid x \leq 8\}$
- ⑦ $(0, \infty)$, $\{x \mid x > 0\}$
- ⑧ $[-3, 10)$, $\{x \mid -3 \leq x < 10\}$
- ⑨ $(-\infty, \frac{3}{8}) \cup (\frac{3}{8}, \infty)$, $\{x \mid x \neq \frac{3}{8}\}$
- ⑩ $[-4, -1]$, $\{x \mid -4 \leq x \leq -1\}$
- ⑪ $[3, 8] \cup (9, 10)$, $\{x \mid 3 \leq x \leq 8 \text{ or } 9 < x < 10\}$
- ⑫ $[-6, 0) \cup [4, 7]$, $\{x \mid -6 \leq x < 0 \text{ or } 4 \leq x \leq 7\}$
- ⑬ $(-\infty, \infty)$. All real numbers
- ⑭ $(-\infty, \infty)$. All real numbers
- ⑮ $(-\infty, \infty)$. All real numbers